

- The mathematical level — the linear algebra and quantum mechanics that explain and justify that line of code.
- The hardware level – the physical processes taking place inside an actual quantum processor when the code is executed.

By examining quantum computing from these three perspectives, you can gain a deeper and more complete understanding of how quantum computers work.

Several important tools and platforms used in quantum computing will be introduced in this series on quantum computing. We will begin with Qiskit, a Python-based, free and open source software development kit (SDK) for quantum computing developed by IBM. We will then move on to OpenQASM (Quantum Assembly Language), an intermediate representation for quantum circuits that enables the low-level description and execution of quantum programs. Next, we will explore Cirq, a free and open source Python framework developed by Google for designing and simulating quantum circuits, with a strong focus on near-term quantum devices. From there, we will proceed to Q#, a domain-specific quantum programming language developed by Microsoft for building quantum algorithms integrated with classical control. Q# is also free and open source and is distributed under the MIT License. In addition, we will become familiar with Qniverse, an indigenously developed quantum computing platform created by C-DAC in India.

There are many misconceptions and myths surrounding quantum computing. As we progress through this series, we will attempt to clarify and dispel several of them. Some of the important questions that will be addressed include the following: Will quantum computers eventually replace classical computers entirely? Does faster code execution in quantum computing depend solely on parallelism, as in GPUs or TPUs? Do qubits allow exponential storage

of information through superposition? Does quantum entanglement enable faster-than-light communication?

Finally, although the focus of the series is quantum computing, we will also briefly explore related areas such as quantum communication and quantum sensing. Topics such as Quantum Key Distribution (QKD) and Post-Quantum Cryptography (PQC) will be discussed in the context of secure communication. In addition, we will touch upon emerging topics in quantum sensing, which exploit quantum phenomena to achieve extremely precise measurements. Together, these discussions will give you a broad and comprehensive understanding of the rapidly evolving landscape of quantum technologies.

A brief history of quantum computing

Let us begin with the scientist most often credited with proposing the idea of quantum computing: Richard Feynman, the brilliant physicist and Nobel laureate. Feynman realised that classical computers cannot efficiently simulate quantum mechanical systems because the computational resources required grow exponentially with the size of the quantum system. In the early 1980s, he proposed that computers based on the principles of quantum mechanics could overcome this limitation. Such machines would exploit uniquely quantum mechanical phenomena such as superposition, entanglement, and interference. Around the same time, the Russian mathematician Yuri Manin independently suggested the possibility of a computing model based on quantum mechanics. For this reason, both Richard Feynman and Yuri Manin are often credited with laying the conceptual foundations of the quantum computer.

Although we will not discuss complex quantum algorithms in detail in this article, it is important to mention two algorithms that ensured the survival and growth of the idea of quantum

computing. While quantum computers were originally proposed to simulate quantum systems, there were initially no practical algorithms demonstrating clear advantages over classical computing. This situation changed dramatically in 1994, when Peter Shor introduced the famous Shor's algorithm.

Shor's algorithm showed that a sufficiently powerful quantum computer could efficiently factor large integers. This result has profound implications for modern cryptography, because many widely used public-key cryptosystems rely on the difficulty of integer factorisation. For example, the RSA cryptosystem, which underlies many secure communications and financial transactions, could be broken if a large-scale quantum computer becomes available. Such a development would have significant consequences for global cybersecurity and the world economy.

Another important milestone came in 1996, when Lov Grover, an Indian computer scientist, introduced Grover's algorithm. This algorithm provides a quadratic speedup for searching an unstructured database, making it significantly faster than classical search methods. Later, it was proven that this quadratic speedup is the best possible for unstructured search, ensuring that Grover's algorithm remains one of the most important works in quantum computing.

Together, these two algorithms demonstrated that quantum computers could have practical applications beyond the simulation of quantum systems. They showed that quantum computing could potentially transform fields such as cryptography, optimisation, and data processing, ensuring that the concept of quantum computing would evolve from a theoretical curiosity into a serious area of research.

Understanding the qubit

Now it is time to begin our technical discussion of quantum computing. The basic unit of quantum information is